Encrypted Smart Complaint Management System

1. Introduction

The Encrypted Smart Complaint Management System is a secure web-based platform designed to allow students and institutional staff to submit, manage, and resolve campus complaints in a confidential and integrity-protected manner. Because complaints often involve sensitive issues such as harassment, misconduct, or administrative failures, strong security controls are essential to protect user identity and complaint data.

This project follows a security-by-design approach, integrating encryption, structured authentication, controlled authorization, and formal threat modeling from the earliest stages of development.

2. Problem Statement

Existing complaint mechanisms often rely on emails, paper forms, or poorly secured portals. These systems commonly lack encryption, structured access control, and proper session security, making them vulnerable to:

- Unauthorized data access
- Identity disclosure
- SQL Injection and XSS attacks
- Session hijacking
- Weak password storage

Given the sensitivity of complaint data, a secure encrypted solution is necessary to guarantee confidentiality, integrity, and accountability.

3. Aim and Objectives

Aim: To design and implement a secure encrypted complaint management system that applies modern secure software engineering practices.

Objectives:

- Implement bcrypt-based password hashing for secure authentication.
- Enforce Role-Based and Rule-Based Access Control.
- Encrypt complaint data at rest using AES-256.
- Secure communication using TLS (HTTPS).
- Prevent common web vulnerabilities (SQL Injection, XSS, CSRF).
- Apply STRIDE-based threat modeling.
- Conduct structured security testing before deployment.

4. System Overview

The system defines three user roles:

- Students can submit complaints (anonymously or identified), track status updates, and view complaint history securely.
- Maintenance Staff can access only assigned complaints and update resolution details.
- Administrators can manage all complaints, control system configurations, generate reports, and manage user roles.

A secure FIFO-based workflow ensures fair complaint handling while preserving strict access boundaries.

5. Secure Architecture and Implementation

The system follows a three-tier architecture:

- Presentation Layer
- Secure forms with CSRF tokens
- Output encoding to prevent XSS
- Server-side validation
- Application Layer
- Authentication and authorization engine
- Encryption/decryption modules
- Secure session management
- Audit logging system
- Data Layer
- PostgreSQL/MySQL database
- AES-256 encrypted complaint storage
- Restricted database privileges
- Security Controls
- Passwords hashed with bcrypt and salted
- TLS 1.2+ for encrypted communication
- Secure HTTPOnly session cookies
- Login attempt limits and session timeout
- Parameterized queries to prevent SQL Injection
- Append-only audit logs for accountability
- Role-based and rule-based authorization enforcing least privilege

6. Threat Modeling and Testing

The system applies the STRIDE model to mitigate:

- Spoofing (strong authentication, TLS)
- Tampering (access control, logging)
- Repudiation (audit trails)
- Information Disclosure (encryption at rest and transit)
- Denial of Service (rate limiting)
- Elevation of Privilege (RBAC enforcement)

Security testing includes static code review, dynamic vulnerability scanning, injection testing, session validation, and encryption verification.

7. Conclusion

The Encrypted Smart Complaint Management System integrates encryption, structured access control, secure authentication, and formal threat modeling into a practical institutional solution. By embedding security throughout the development lifecycle, the system ensures confidentiality, integrity, accountability, and resilience against modern cyber threats.